# The Last PhD We Will Ever Award: Soundness Limits of the Viva Protocol Under Large Language Model Oracles

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**Abstract**

Doctoral degrees – particularly in computer science – have long relied on a simple protocol: a candidate produces a dissertation and a committee interrogates the candidate in a viva voce defense. The defense is meant to certify a latent property (independent research capability) using only a bounded interaction. We model this *Viva Protocol* as an interactive proof system in the sense of [12], and we treat modern large language models (LLMs) as cheap *oracles* available to the prover. This perspective reduces contemporary anxiety about generative text and "AI-written dissertations" to a classical question about verifier resources: how much interaction, scrutiny, and replication are required to preserve soundness when the prover can cheaply generate fluent, context-dependent transcripts? We provide (i) a formalization of doctoral examination as an interactive proof with an unobservable statement, (ii) a soundness degradation theorem and a stronger conjecture giving conditions under which soundness cannot be simultaneously maintained with fairness and bounded verifier cost, (iii) a threat model for LLM-oracle provers, (iv) a simulation framework together with a concrete Monte Carlo stress test, and (v) a systematic analysis of "patches" (air-gapping, provenance, detection, watermarking, replication) and why they fail under adaptive attackers or institutional incentives. We conclude with an incident-report-style narrative of the "last" PhD – the final award before the protocol becomes, in equilibrium, more ceremonial than certifying – and discuss replacement credentialing mechanisms.

## Executive Summary

The viva voce defense is intended to establish the truth of a latent statement: that the candidate can conduct independent research. In practice it uses bounded interaction, partial artifact inspection, and committee judgment. In this paper we treat the defense as a protocol with classical completeness and soundness desiderata [12, 24], and we ask what happens when the prover has access to an LLM oracle.

Our core claim is conditional but sharp: once oracle-assisted candidates can produce defense transcripts that are sufficiently indistinguishable from those of genuinely competent candidates, the viva loses soundness unless committees expand verification budgets or accept fairness costs. This mirrors patterns in adversarial ML: defenses that work against non-adaptive attackers often fail once the attacker optimizes to the detector [6, 28]. Empirically, the relevant phenomenon is already visible in adjacent assessment regimes: contract cheating is industrialized [10, 22], and "authentic assessment" is not, by itself, a guarantee of integrity [9].

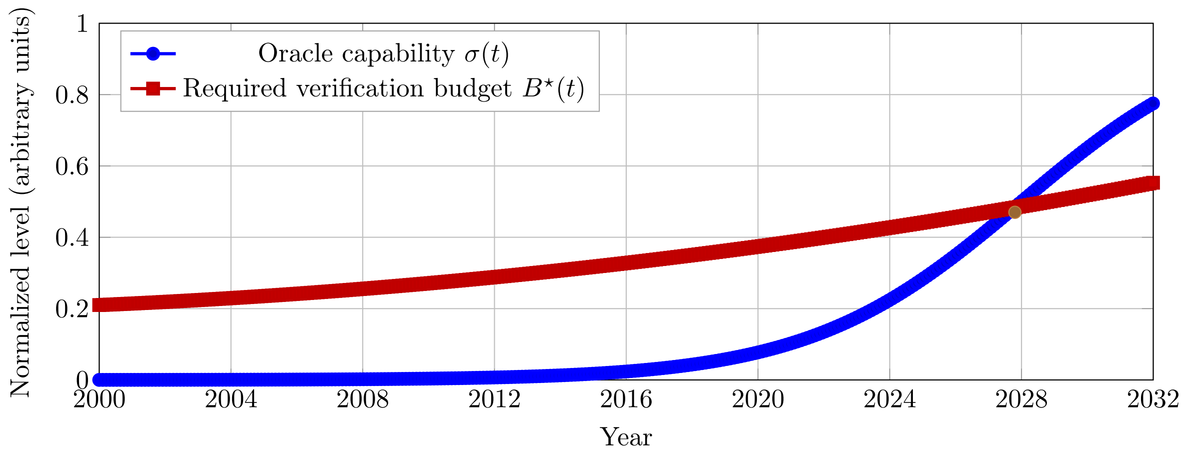


Figure 1: A schematic threshold curve. The blue line is a normalized oracle-capability level $\sigma(t)$; the red line is the minimum verification budget $B^\star(t)$ needed to preserve a target soundness level. The brown point marks the illustrative future crossover at which the normalized curves meet. This figure is conceptual and parameterized; Section 5 explains how the simulation operationalizes the same tension.

We contribute: (1) a formal model of the Viva Protocol (Section 3); (2) a soundness degradation theorem plus a soundness–fairness–cost conjecture (Section 4); (3) a threat model for LLM-oracle provers grounded in the LLM evaluation literature (Section 4); (4) a simulation framework aligned with a concrete Monte Carlo instantiation (Sections 5–6); (5) a patch catalog with formal and empirical reasons for failure (Section 7); (6) an incident report narrative of the "last PhD" (Section 8); and (7) a plan for replacing the PhD with credentials that certify something institutionally verifiable (Section 9).

# 1 Introduction

The doctoral degree is simultaneously (i) a training pipeline, (ii) a labor-market signal under information asymmetry [1, 25], (iii) a screening or filtering device [2, 27], and (iv) a community mechanism for assigning epistemic responsibility: someone must be accountable for a dissertation's claims.

The viva voce defense is the keystone ceremony and keystone check. It is meant to certify properties that are hard to observe directly: independence, depth, robustness of understanding, and the ability to respond to critique. However, large language models change the economics of producing *plausible scholarly artifacts* and *plausible scholarly discourse*. Scaling and instruction-following yield models capable of fluent responses on broad tasks [5, 17], and prompting techniques elicit longer reasoning traces that can appear like derivations [33]. At the same time, multiple lines of work show that (i) models can produce false but compelling statements [18], (ii) hallucinations can be hard to detect without external checks [19], and (iii) even evaluators (humans or LLM judges) exhibit systematic biases and vulnerabilities to perturbations [7, 14, 34].

This paper treats the viva as a protocol whose security properties can be analyzed. The question is not whether cheating exists in academia; it does, and it already supports an industry [10, 22]. The question is what *remains verifiable* at reasonable cost and with acceptable fairness once fluent transcript generation becomes cheap.

The formal and empirical parts of the paper track the same question. The formal side says that if an oracle-assisted candidate can induce nearly the same transcript law as a competent candidate, then completeness for the latter implies dangerous acceptance probability for the former. The empirical side asks how much committee structure, follow-up pressure, and post-defense verification are needed to widen that gap in practice.

## 2 Related Work

We situate our analysis across four bodies of primary literature.

**Interactive proofs and verifier-resource tradeoffs.** Interactive proof systems were introduced by [12] with probabilistic verification via interaction. Subsequent work analyzes the power of interaction and randomness through Arthur–Merlin games and related protocols [3,4,13], culminating famously in IP = PSPACE [24]. Our contribution is not new complexity theory but a mapping: the viva can be analyzed as an interactive proof where verifier resources are committee time, expertise, and willingness to replicate.

**Credentialing, signaling, and screening.** Degrees operate in markets with information asymmetry [1], and are often modeled as signals [25] or filters [2]. If a signal becomes cheaper to produce (or easier to simulate), equilibria shift toward either signal inflation or new forms of screening [27]. Our protocol analysis can be read as a mechanically precise version of this shift: LLMs reduce the marginal cost of producing a "plausible dissertation transcript", so the committee must either spend more to verify or accept weaker semantics.

**Human evaluation, LLM evaluation, and LLM-as-a-judge.** Large-scale evaluation frameworks emphasize that accuracy alone is insufficient, and that robustness and fairness trade off against each other [17]. Task collections like BIG-bench broaden coverage and show that capability varies across tasks in complicated ways [26]. In open-ended settings, the idea of using LLMs as judges has been studied systematically; these works note biases (position, verbosity, self-enhancement) and propose mitigations [34], while others document that both human and LLM judges are vulnerable to perturbations and adversarial attacks [7]. Surveys emphasize reliability challenges and motivate careful design [14].

**Adversarial ML, detection, watermarking, and academic integrity.** Adversarial examples and adaptive attackers are a standard warning: defenses that appear to work can fail under new attacks [6, 28]. Text detection tools can be helpful but face false-positive risk and evasion [11, 20]. Watermarking seeks detectability at generation time [16] but faces paraphrase and reverse-engineering attacks [23]. In education, contract cheating is industrialized [10, 22], and evidence suggests that assessment design changes alone do not eliminate it [9].

## 3 A Formal Model of the Viva Protocol

We model a viva defense as an interactive proof system whose statement is not directly observable.

### 3.1 Protocol objects and transcript

**Definition 1** (Question, answer, transcript)**.** Let $\mathcal{Q}$ be a space of committee questions (including follow-ups), $\mathcal{A}$ a space of candidate answers (spoken text, whiteboard derivations, code, demonstrations), and $\mathcal{T} = (\mathcal{Q} \times \mathcal{A})^*$ the space of finite transcripts. A transcript $t \in \mathcal{T}$ is a sequence $((q_1, a_1), \ldots, (q_n, a_n))$.

**Definition 2** (Viva Protocol)**.** Let $\mathcal{H}$ be the population of candidates. Each candidate $h \in \mathcal{H}$ induces a possibly randomized answering strategy $P_h$; if the candidate has access to an oracle, we write the resulting strategy as $P_h^{\mathcal{O}}$. A *Viva Protocol* consists of a verifier $\mathsf{V}$ and the family of candidate-dependent prover strategies $\{P_h : h \in \mathcal{H}\}$. The committee $\mathsf{V}$ adaptively chooses each question as a function of the current transcript and any external verification it performs, and then halts with decision $\mathsf{V}(t) \in \{\text{accept}, \text{reject}\}$.

For a fixed candidate $h$, we write $\Pr[\mathsf{V} \leftrightarrow P_h \Rightarrow \text{accept}]$ for the acceptance probability induced by internal randomness of $\mathsf{V}$, of $P_h$, and of any verification procedures.

This resembles the standard interactive-proof setting [12], except the "instance" is not a string $x$ but a sociotechnical bundle (dissertation, artifacts, prior work, and the candidate)[1]. The distinction between $h$ and $P_h$ is important: $h$ is the underlying person and latent state; $P_h$ is the observable defense behavior that the protocol sees.

### 3.2 Latent competence as counterfactual robustness

To formalize the claim being verified, we define competence operationally via *counterfactual robustness*: the ability to handle perturbations, error cases, and extensions.

**Definition 3** (Task distribution and perturbations)**.** Let $\mathcal{D}$ be a distribution over base questions $q \in \mathcal{Q}$. Let $\mathcal{R}$ be a set of perturbation operators $r : \mathcal{Q} \to \mathcal{Q}$ that represent "nearby" questions: changed assumptions, adversarial edge cases, alternative definitions, modified experimental setups, or alternate architectures.[2]

**Definition 4** (Unassisted response function)**.** Each candidate $h \in \mathcal{H}$ has an implicit *unassisted* response function $f_h : \mathcal{Q} \to \Delta(\mathcal{A})$ mapping a question to a distribution over answers given the candidate's internal knowledge and skills. Assume there exists a possibly expensive scoring predicate $\mathsf{Correct}(q, a) \in \{0, 1\}$ capturing whether answer $a$ appropriately addresses question $q$. In practice, the committee only approximates $\mathsf{Correct}$ under a finite verification budget.

**Definition 5** ($(\epsilon, \rho)$-robust competence)**.** Fix $\epsilon, \rho \in [0, 1]$. A candidate $h$ is $(\epsilon, \rho)$*-robustly competent* under $(\mathcal{D}, \mathcal{R})$ if

$$\Pr_{q \sim \mathcal{D},\, r \sim \mathrm{Unif}(\mathcal{R}),\, a \sim f_h(r(q))} \big[\mathsf{Correct}(r(q), a) = 1\big] \geq 1 - \epsilon,$$

and, additionally, the candidate maintains this accuracy under a $\rho$-fraction of adaptive perturbations chosen after observing previous answers.

The emphasis on robustness is motivated by the evaluation literature: small in-distribution samples can overestimate capability when respondents exploit shallow cues or when evaluators are biased [7, 17].

[1] In computer science, this bundle often includes a PDF, a directory tree, three unversioned scripts, and a README whose first imperative verb is "just."

[2] A separate perturbation family concerns institutional phenomena such as adviser turnover or unavailable infrastructure. We omit those.

### 3.3 Completeness and soundness desiderata

We map doctoral ideals onto proof-system properties.

**Definition 6** (Completeness and soundness)**.** Fix a target competence class $\mathsf{Comp} \subseteq \mathcal{H}$. A Viva Protocol is *$\alpha$-complete* if for every $h \in \mathsf{Comp}$,

$$\Pr[\mathsf{V} \leftrightarrow P_h \Rightarrow \text{accept}] \geq 1 - \alpha.$$

It is *$\beta$-sound* if for every $h \notin \mathsf{Comp}$,

$$\Pr[\mathsf{V} \leftrightarrow P_h \Rightarrow \text{accept}] \leq \beta.$$

In real departments, $\mathsf{Comp}$ informally means "can do independent research". The formal model makes explicit the central difficulty: the system must certify a property of the candidate's internal capability, but it observes only a bounded transcript and a limited amount of external checking.

## 4 LLM-Oracle Provers and an Impossibility Result

### 4.1 Threat model: access regimes for LLM oracles

We treat an LLM as an oracle $\mathcal{O}$ mapping prompts to distributions over responses. Because LLM outputs can be fluent but unfaithful [18] and self-consistency is not equivalent to truth [19], oracle access changes the transcript distribution without necessarily improving unassisted robustness.

**Definition 7** (LLM oracle)**.** An *LLM oracle* is a stochastic map $\mathcal{O} : \mathcal{P} \to \Delta(\mathcal{A})$ from prompts $\mathcal{P}$ (which may include the dissertation, slides, prior Q&A, and committee identities) to answers.

**Definition 8** (Oracle access regimes)**.** We consider three regimes for a candidate strategy with oracle access:

1. **Preparation-only oracle:** the candidate queries $\mathcal{O}$ before the defense to produce documents and rehearse likely exchanges.
2. **Live oracle (covert):** the candidate queries $\mathcal{O}$ during the viva with bounded latency by hidden device or second screen.
3. **Live oracle (permitted):** tool use is allowed and disclosed; the verified statement shifts from "unassisted competence" to "tool-mediated performance".

**Adversary goal.** The adversary seeks *false accept*: passing the viva while failing the target unassisted competence predicate. This is analogous to evasion against a detector: the objective is to cross the verifier's decision boundary, not to be correct.[3]

### 4.2 A transcript indistinguishability lemma

We start with the standard observation that if two candidates induce similar transcript laws under the same verifier, then no acceptance rule based only on those transcripts can separate them well.

**Definition 9** (Transcript distribution)**.** Fix a verifier $\mathsf{V}$ and a candidate strategy $P$. The interaction $\mathsf{V} \leftrightarrow P$ induces a distribution over transcripts $t \in \mathcal{T}$, denoted $\mathsf{Trans}(\mathsf{V}, P)$.

[3] When rigorous verification is absent, correctness becomes an optional feature.

**Definition 10** (Total variation distance)**.** For two probability measures $\mu$ and $\nu$ on the same transcript space,

$$\mathrm{TV}(\mu, \nu) := \sup_{E \subseteq \mathcal{T}} |\mu(E) - \nu(E)|.$$

For discrete transcript spaces this equals $\frac{1}{2} \sum_{t \in \mathcal{T}} |\mu(t) - \nu(t)|$.

**Lemma 1** (Acceptance gap bounded by total variation)**.** *For any verifier* $\mathsf{V}$ *and any two candidate strategies* $P_1, P_2$*,*

$$\big| \Pr[\mathsf{V} \leftrightarrow P_1 \Rightarrow \text{accept}] - \Pr[\mathsf{V} \leftrightarrow P_2 \Rightarrow \text{accept}] \big| \leq \mathrm{TV}\big(\mathsf{Trans}(\mathsf{V}, P_1), \mathsf{Trans}(\mathsf{V}, P_2)\big).$$

**Intuition.** The acceptance decision is just an indicator function on transcripts. Total variation is the largest possible difference that any such event can have under two distributions. So if two defense transcript laws are close in total variation, every committee rule operating on those transcripts must assign nearly the same acceptance probability to both.

### 4.3 A soundness degradation theorem

We isolate the structural failure mode: if an oracle-assisted candidate can emulate the transcript law of a competent candidate under a given committee strategy, soundness collapses.

**Assumption 1** (Committee resource constraints)**.** The committee $\mathsf{V}$ has a bounded interaction budget of at most $T$ question-answer rounds and a bounded verification budget $B$ (time, compute, and attention). In particular, $\mathsf{V}$ cannot evaluate $\mathsf{Correct}(q, a)$ exhaustively for every exchange and must rely on selective checking, partial replication, or heuristic scoring.[4]

**Assumption 2** (Oracle transcript emulation)**.** Fix a competent candidate $h^+ \in \mathsf{Comp}$ and an incompetent candidate $h^- \notin \mathsf{Comp}$. There exists an oracle-assisted strategy $P_{h^-}^{\mathcal{O},\mathrm{em}}$, where em denotes *emulation* – i.e., a strategy chosen to make the induced transcript distribution mimic that of a competent candidate under $\mathsf{V}$ – such that

$$\delta_{\mathrm{tr}}(\mathsf{V}; h^+, h^-) := \mathrm{TV}\big(\mathsf{Trans}(\mathsf{V}, P_{h^+}), \mathsf{Trans}(\mathsf{V}, P_{h^-}^{\mathcal{O},\mathrm{em}})\big) \leq \eta$$

for some $\eta \in [0, 1]$. Here $\eta$ is an upper bound on transcript distinguishability: small $\eta$ means the oracle-assisted defense looks very similar to the competent one from the committee's point of view.

**Theorem 1** (Viva soundness bound under oracle emulation)**.** *Under Assumptions 1 and 2, any Viva Protocol that is* $\alpha$*-complete for* $h^+$ *satisfies*

$$\Pr[\mathsf{V} \leftrightarrow P_{h^-}^{\mathcal{O},\mathrm{em}} \Rightarrow \text{accept}] \geq 1 - \alpha - \eta.$$

*Equivalently, when transcript distinguishability is at most* $\eta$*, no committee rule that preserves completeness for the competent candidate can push the false-accept probability for the emulated candidate below* $1 - \alpha - \eta$*.*

**Proof sketch.** By completeness, $\Pr[\mathsf{V} \leftrightarrow P_{h^+} \Rightarrow \text{accept}] \geq 1 - \alpha$. By Lemma 1, replacing $P_{h^+}$ with $P_{h^-}^{\mathcal{O},\mathrm{em}}$ changes acceptance probability by at most $\eta$. □

[4] We assume committee members are human and therefore experience fatigue. Future work may relax this assumption.

**Interpretation.** Theorem 1 says that soundness is controlled by transcript distinguishability. LLMs are a technology for shrinking it along the cues that committees often use informally: fluency, calmness, plausible citations, and locally coherent explanations. Unless committees spend budget $B$ on direct correctness checks, or intentionally choose question types that make oracle-emulated and human-generated defenses diverge, the protocol drifts toward ceremonial acceptance.

### 4.4 A tri-lemma conjecture: soundness, fairness, and cost

The natural response is to try to *increase* transcript distinguishability – by harder perturbations, more replication, or stronger provenance requirements. Each move has fairness and cost consequences.

**Conjecture 1** (Soundness–fairness–cost tri-lemma)**.** *Fix a target competence class* Comp *and a heterogeneous candidate population in which performance under stress, language conditions, and accommodations is unequal. For any Viva Protocol with bounded verifier resources* $(T, B)$*, there exists a threshold oracle capability* $\sigma^\star$ *such that for all* $\sigma \geq \sigma^\star$ *one cannot simultaneously:*

*(i) keep false accepts under oracle attack small (soundness),*
*(ii) keep false rejects low across subpopulations under humane conditions (fairness), and*
*(iii) keep verifier resources bounded (cost).*

**Evidence base.** The conjecture is motivated by (a) empirical evidence that open-ended evaluation is prone to judge bias and adversarial vulnerabilities [7, 34], (b) institutional guidance that detection tools have false positives and should not be treated as decisive evidence [30], and (c) the standard adversarial lesson that published defenses invite adaptation [6, 28].

## 5 Simulation Framework

Sections 5 and 6 are deliberately aligned. The implemented study is narrower than the full design space from Section 4: it does *not* separately sweep committee size, total meeting length, and post-defense budget. Instead it bundles those effects into four protocol families that correspond to familiar viva styles. This keeps the simulation readable, reproducible, and faithful to what Section 6 actually reports.

### 5.1 Question families

Every simulated exchange belongs to one of four question families. Table 1 gives the operational definitions used throughout the paper.

These families are chosen because they separate rehearsable discourse from on-the-fly repair. In particular, the distinction between perturb and debug matters: the former asks whether a result generalizes when assumptions are changed; the latter asks the candidate to find and fix a concrete local failure.

### 5.2 Committee protocols

The simulation instantiates four committee protocols. Table 2 summarizes them and is the definitive bridge between the framework and the Monte Carlo study.

The protocols absorb multiple real-world choices at once: how much the committee values correctness over performance style, how often it catches unsupported claims, and whether it spends scarce time on audits rather than oral questioning. In that sense, each row of Table 2 should be read as a coarse policy bundle rather than a single literal departmental rule.

| Family | Meaning in the viva | Why oracle assistance matters differently |
|---|---|---|
| Stock | Routine background questions, standard definitions, or rehearsable justifications of the dissertation's main claims | Strongest opportunity for memorization, rehearsal, and canned fluency |
| Method | Questions about experimental design, proof structure, modeling choices, or implementation decisions | Tool assistance helps summarize and defend familiar choices, but less reliably than for stock questions |
| Perturbation | The committee changes an assumption, poses an edge case, or asks whether the result survives a nearby modification | Tests counterfactual robustness rather than memorized exposition |
| Debug | The committee presents a broken derivation, failed run, inconsistency, or counterexample and asks the candidate to diagnose or repair it | Forces local reasoning under pressure; here the candidate is the one debugging |

Table 1: Question families used in the simulation. The labels `stock`, `method`, `perturb`, and `debug` match the code in Appendix A.

### 5.3 Candidate groups and latent variables

We simulate three candidate groups.

1. **Human-only**: strong latent knowledge, moderate fluency, and nontrivial oral-performance vulnerability.

2. **Human+LLM**: the same underlying knowledge distribution as the human-only group, but with better fluency and better performance on stock and method questions, representing drafting and rehearsal assistance.

3. **LLM-front**: high discursive fluency and good stock performance, but weak perturbation and debug robustness together with a higher tendency to make fluent unsupported claims.

Each candidate $i$ carries three latent variables: knowledge $k_i$, discourse fluency $\phi_i$, and oral-performance vulnerability $a_i$. Let $g(i)$ denote the candidate's group label (e.g., human, human+LLM, or LLM-front). The group label governs both the distribution of latent variables (via group-specific means) and question-family bonuses $b_{g(i),\tau}$, where $\tau$ indexes a question type (e.g., stock, method, perturb, debug); these bonuses capture systematic advantages or weaknesses of that group on different types of questions.

### 5.4 Correctness, fluency, and committee-side scoring

The simulation separates *being right* from *sounding right*. That separation is intentional: examiner narratives and judge-bias studies suggest that oral credibility and substantive mastery are correlated but not identical [7, 14, 29].

For candidate $i$, question instance $j$, question family $\tau$, and committee protocol $s$, latent correctness is generated by the item-response-style model

$$\Pr[y_{ij\tau} = 1] = \text{logistic}\big(k_i + b_{g(i),\tau} - d_{j\tau} - \lambda_s a_i \rho_\tau\big),$$

| Protocol | Question mix (stock, method, perturb, debug) | Interpretation |
|---|---|---|
| Conventional | $(5, 3, 2, 2)$ | Ordinary defense emphasizing comprehension and contribution, with modest follow-up pressure |
| Structured | $(3, 3, 3, 3)$ | More even coverage and tighter rubric; lower committee noise and more systematic catching of unsupported claims |
| Adversarial | $(1, 3, 4, 4)$ | Heavy emphasis on perturbation and debugging; strongest pressure on transcript distinguishability, but also the harshest stress profile |
| Replication-heavy | $(2, 2, 2, 2)$ plus two artifact audits | Fewer oral questions, with effort shifted toward code, proof, or artifact checking |

Table 2: Committee protocols used in the simulation. The prose discussion in this section and the numerical study in Section 6 refer to these same four protocols.

where $d_{j\tau}$ is question difficulty, $\rho_\tau$ is the stressfulness of family $\tau$, and $\lambda_s$ is the stress multiplier associated with protocol $s$. This is a deliberately minimal logistic specification: more knowledge helps, harder questions hurt, and stress harms performance most on the families meant to probe robustness. It should be read as a stylized item-response model, not as a claim that real vivas obey a literal logistic law.

The committee does not observe $y_{ij\tau}$ directly. Instead it observes a real-valued per-question score

$$
\begin{aligned}
u_{ij\tau} = {} & w_s^{(c)} y_{ij\tau} + w_s^{(f)} F_{i\tau} + \varepsilon_{ij\tau} \\
& + \alpha\, \mathbf{1}\{z_{ij\tau} = 1 \wedge c_{ij\tau} = 0\} - \beta\, \mathbf{1}\{c_{ij\tau} = 1\},
\end{aligned}
$$

where $F_{i\tau} = \text{logistic}(\phi_i + \delta_\tau) \in (0, 1)$ is the committee's impression of fluency, $z_{ij\tau}$ is an unsupported but fluent claim (a "slip"), $c_{ij\tau}$ records whether the committee catches that slip, and $\varepsilon_{ij\tau}$ is committee-side scoring noise. Here "committee-side scoring noise" means nothing mysterious: it is simply unexplained variation in how the committee interprets the same exchange. The weights $w_s^{(c)}$ and $w_s^{(f)}$ specify how much a protocol rewards correctness versus fluency.

Table 3 summarizes the symbols that matter most for reading the model.

**Why this model?** The correctness equation is the smallest model we know that captures the paper's substantive distinction: stock and method questions reward preparation, while perturbation and debug questions expose whether the candidate can still reason when the script breaks. The committee-side score equation is equally intentional. It lets fluency help when no verification occurs, lets caught unsupported claims hurt, and permits structured committees to be both less noisy and more likely to catch slips.

### 5.5 Verification model and capability sweep

Verification is not modeled as an omniscient oracle. Instead, protocols differ in how likely they are to catch unsupported claims and in whether they perform audits. The replication-heavy committee

| Symbol | Meaning |
|---|---|
| $k_i$ | latent knowledge of candidate $i$ |
| $\phi_i$ | discourse fluency of candidate $i$ |
| $a_i$ | oral-performance vulnerability of candidate $i$ |
| $d_{j\tau}$ | difficulty of question $j$ in family $\tau$ |
| $\rho_\tau$ | stressfulness of question family $\tau$ |
| $\lambda_s$ | protocol-specific stress multiplier |
| $y_{ij\tau}$ | latent correctness indicator for that exchange |
| $F_{i\tau}$ | perceived fluency for candidate $i$ on family $\tau$ |
| $u_{ij\tau} \in \mathbb{R}$ | committee-side score before averaging across the defense |
| $w_s^{(c)}, w_s^{(f)}$ | protocol-specific weights on correctness and fluency |
| $z_{ij\tau}, c_{ij\tau}$ | unsupported claim indicator and whether the committee catches it |

Table 3: Glossary for the simulation model. The equations in this section and the code in Appendix A use the same conceptual variables, although the code stores many of them as arrays or named parameters.

performs two artifact audits, which can penalize internally inconsistent written records. Detection-like signals are therefore *advisory*: they change scores or trigger follow-up, but no single suspicious cue automatically fails a candidate. This is the sense in which the simulation encodes operational constraints.

The implemented study also does not estimate oracle capability from an external benchmark suite. Instead it treats oracle strength operationally. The baseline LLM-front group fixes one capability level; the sensitivity study in Section 6 then scales that group by a single multiplier $\gamma \in [0.7, 1.3]$, improving its stock and method performance and modestly lowering its falsehood rate. This is narrower than a full empirical calibration of $\sigma$, but it matches the code exactly and suffices to show how protocol rankings move as the adversary gets stronger.

A limitation of the framework is worth stating explicitly. The model separates knowledge $k_i$ from fluency $\phi_i$ and only couples them through coarse group-level means. Real candidates coordinate those traits: preparation, insight, confidence, and speaking skill often rise together. The simulation intentionally breaks that coordination so that one can isolate the part of committee judgment that is vulnerable to fluent but shallow performance.

# 6 Simulation study

The goal of the simulation is not to estimate any real university's pass rate. It is a comparative stress test of doctoral examination protocols under oracle-assisted answering.

## 6.1 Experimental setup

All reported numbers come from a fixed-seed Monte Carlo study with 50,000 candidates in each candidate-group/protocol cell, for a total of 600,000 simulated defenses. The implementation uses only `numpy`, `pandas`, and `matplotlib`, requires no API access, and generates the summary tables and sensitivity outputs from scratch; the full script appears in Appendix A.

To measure the gap between performance and understanding, each candidate also receives a *hidden robustness score*: mean accuracy on 20 withheld perturbation and debug questions that are never shown to the committee and are scored without any fluency term. This operationalizes the

| Protocol | Human-only | Human+LLM | LLM-front |
|---|---|---|---|
| Conventional | 75.7 | 88.2 | 28.0 |
| Structured | 70.1 | 81.1 | 3.5 |
| Adversarial | 57.4 | 69.2 | 0.8 |
| Replication-heavy | 65.3 | 73.5 | 4.9 |

Table 4: Pass rates (%) by candidate group and committee protocol. Each cell uses 50,000 simulated candidates.

counterfactual robustness notion from Section 3. The absolute pass rates should therefore be read as model outputs, not institutional facts; the comparative quantities are the object of interest.

### 6.2 Pass rates and the soundness–fairness frontier

Table 4 reports pass rates by candidate group and protocol. The conventional committee is visibly unsound: it passes 28.0% of LLM-front candidates. Tightening the protocol helps, but only by sacrificing fairness to human-only candidates. Structured questioning reduces the LLM-front pass rate to 3.5%, replication-heavy examination to 4.9%, and adversarial questioning to 0.8%. The corresponding pass rate for human-only candidates falls from 75.7% under the conventional committee to 70.1% (structured), 65.3% (replication-heavy), and 57.4% (adversarial). The human+LLM group dominates the human-only baseline under every protocol, which is unsurprising in a world where drafting, rehearsal, and literature synthesis can be partly off-loaded to a model.

Because each cell contains 50,000 simulated defenses, uncertainty on pass rates is small. For example, the conventional committee passes human-only candidates at 75.7% (Wilson score 95% CI $[75.3, 76.1]$) and LLM-front candidates at 28.0% (Wilson score 95% CI $[27.6, 28.4]$). The qualitative ordering is unchanged across random seeds and across the capability sweep reported below.

Figure 2 makes the central tradeoff explicit. Relative to the conventional committee, structured questioning cuts LLM-front false accepts by roughly 87.6% but increases false rejects on human-only candidates from 24.3% to 29.9%. Adversarial questioning cuts false accepts by roughly 97.0%, but false rejects rise to 42.6%. Replication-heavy examination sits between those extremes.

### 6.3 Confidence–robustness gap and capability sensitivity

The deeper problem is not merely false acceptance; it is false acceptance with confidence. Table 5 compares committee confidence among passers with their hidden robustness. Under the conventional committee, human-only passers have mean confidence 0.740 and hidden robustness 0.698. LLM-front passers, by contrast, have nearly the same mean confidence (0.715) but drastically lower hidden robustness (0.162). Across all four protocols, passing LLM-front candidates have committee confidence between 0.681 and 0.715, while their hidden robustness lies only between 0.162 and 0.193. The protocol is therefore not only leaky; it is overconfident about its leaks.

Finally, we perform a one-parameter capability sweep in which only the LLM-front group is strengthened, holding the committee protocols fixed. Figure 3 shows that the conventional committee degrades fastest: as the LLM-front capability multiplier $\gamma$ rises from 0.7 to 1.3, its pass rate increases from 20.1% to 36.7%. Over the same range, the structured, replication-heavy, and adversarial protocols rise from 1.9% to 6.7%, 2.4% to 7.9%, and 0.3% to 1.8%, respectively. The ordering is stable, but so is the frontier: protocols that are harder to fool are also harsher on human candidates.

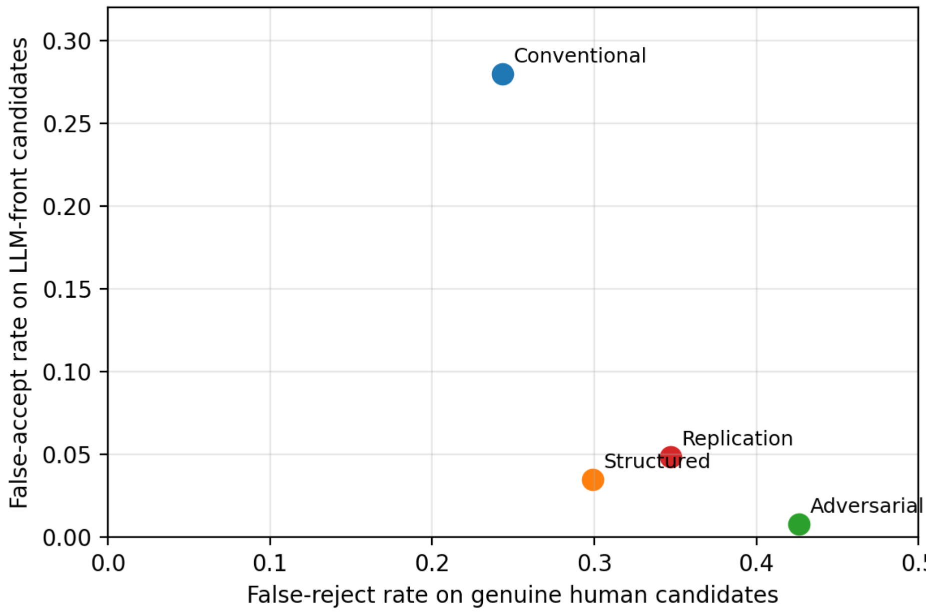


Figure 2: Soundness-fairness frontier induced by the four committee protocols. Moving downward improves soundness against LLM-front candidates; moving left reduces false rejects on genuine human candidates. No protocol dominates.

| Protocol | Human conf. | Human robust. | LLM conf. | LLM robust. |
|---|---|---|---|---|
| Conventional | 0.740 | 0.698 | 0.715 | 0.162 |
| Structured | 0.727 | 0.708 | 0.687 | 0.183 |
| Adversarial | 0.723 | 0.718 | 0.681 | 0.193 |
| Replication-heavy | 0.749 | 0.706 | 0.711 | 0.173 |

Table 5: Mean committee confidence and mean hidden robustness among *passing* candidates. Confidence is the committee-side acceptance score mapped to $[0, 1]$; robustness is withheld perturbation/debug accuracy.

## 6.4 Threats to validity and reproducibility

This section is a protocol-level stress test, not an empirical audit of any institution. The candidate groups, question mixes, and scoring rules are stylized; the absolute pass rates are therefore model outputs, not measured institutional facts. We do not model field-specific novelty, adviser influence, prior publications, or covert live tool use during the defense.

One limitation deserves special emphasis. The simulation separates knowledge from fluency much more sharply than most real departments do. In practice, strong candidates often become fluent precisely because they understand the material well, while weaker candidates may compensate by over-rehearsing stock discourse. The model intentionally exaggerates that separation so that the protocol's vulnerability to style-sensitive judgment is measurable. The resulting numbers should be read comparatively, not literally.

Reproducibility is deliberately straightforward[5]. The complete source code used to generate the

[5] This places the appendix in the unusual position of being easier to verify than the credentialing protocol it simulates.

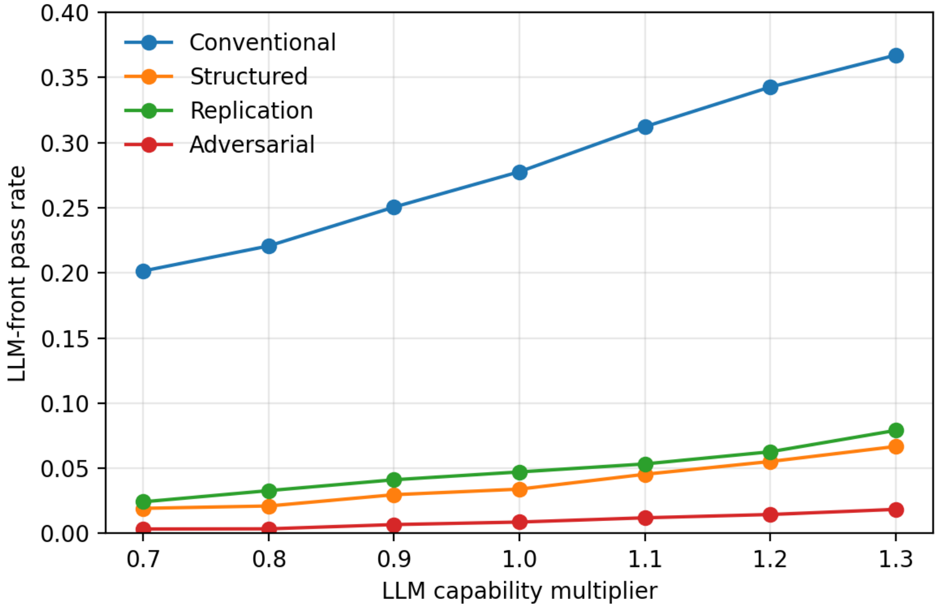


Figure 3: Sensitivity of LLM-front pass rates to a scalar capability multiplier. The conventional committee degrades fastest as the oracle-assisted candidate becomes stronger.

simulation appears in Appendix A. It fits in a short Python script, uses a fixed random seed, and requires no external dataset, online service, or nonstandard infrastructure.

## 7 Patches and Why They Fail

We analyze countermeasures as security patches under adaptive attackers and budget constraints. Table 6 gives the high-level picture; the subsections justify the entries.

### 7.1 Air-gapping and the preparation-only oracle

Air-gapping targets the live-oracle channel. However, preparation-only oracle access is sufficient to generate rehearsed responses and to engineer artifacts that survive low-budget inspection. Air-gapping therefore increases transcript distinguishability only for one attack surface. It also shifts the protocol toward "performance under pressure", which risks conflating research competence with stress tolerance and language performance. In values-based integrity frameworks, fairness and respect are central considerations [15].

### 7.2 Provenance, attestation, and compliance theater

Cryptographic provenance standards aim to attach verifiable metadata to content [8] and to issue machine-verifiable credentials [32]. Major AI providers discuss provenance methods and watermarking as part of an integrity toolkit [21]. However, provenance proves the history of an artifact, not the understanding of a person, and provenance ecosystems are brittle: they require consistent tool support and can be disrupted by benign transformations.

More subtly, provenance requirements invite *compliance theater*: candidates optimize for satisfying

| Patch | Soundness gain | Fairness risk | Failure mode under adaptation |
|---|---|---|---|
| Air-gapped defense | Medium (against live oracle) | Medium–High | Preparation-only oracle remains; screening shifts toward stress tolerance |
| Provenance logs | Medium (for artifacts) | Low–Medium | Logs can be synthesized, stripped, or bypassed; checklist compliance replaces understanding |
| Text detection | Low–Medium | High | False positives, evasion, and bypasser tools; detectors remain advisory rather than dispositive [30] |
| Watermarking | Medium (when intact) | Medium | Paraphrasing and reverse engineering erase or evade the signal [23] |
| Replication requirement | High (when actually done) | Low–Medium | Strongest patch, but expensive and therefore undersupplied |

Table 6: Qualitative patch matrix. "Soundness gain" means expected improvement in transcript distinguishability and corresponding reduction in false accepts, holding broad institutional conditions fixed.

the provenance checklist, including synthetic but plausible logs. This is structurally similar to log-farming failures in security compliance.

### 7.3 Detection: assistance, not adjudication

GLTR explicitly frames detection as support for human judgment through statistical cues [11], and DetectGPT proposes a model-aware technique based on probability curvature [20]. Yet in institutional settings false positives are high-stakes and unevenly distributed. Commercial guidance explicitly warns that AI-writing detection may misidentify text and should not be used as the sole basis for adverse action [30]. Release notes show an arms race including explicit detection of AI paraphrasing and bypasser tools [31].

In the language of Section 4, detectors help only insofar as committees are willing to let them change decisions. If detector outputs remain advisory, they may trigger more questions or a modest audit, but they do not by themselves restore soundness.

### 7.4 Watermarking and adaptive attacks

Watermarking modifies generation to embed a detectable signal [16]. In the idealized case, watermarking can increase transcript distinguishability by making assistance more detectable. However, adaptive attacks can paraphrase or reverse-engineer the scheme; EMNLP results show that *even* limited access to a watermarked model can enable more effective paraphrasing attacks that evade detection [23]. This is precisely the adaptive-attacker dynamic familiar from adversarial ML [6, 28].

### 7.5 Replication: the only strong patch, and why it is scarce

Replication and re-derivation come closest to a cryptographic check on competence: they ask whether the candidate can actually reproduce or extend the result when the script breaks. When performed, they directly increase transcript distinguishability between robust understanding and merely fluent defense performance. But replication is expensive, time-consuming, and infrastructure-dependent. In academia, where committees face strong opportunity costs, that patch is chronically under-provided [9, 22].

## 8 Incident Postmortem: The Last PhD

We present a postmortem narrative in the style of an engineering incident report. The narrative is fictionalized but intended as a realistic design probe.[6]

### Summary

On 2026-05-14 (local time), the Department awarded a doctorate that was later recognized (by consensus, not statute[7]) as the last PhD for which the viva retained nontrivial soundness against LLM-oracle provers. Subsequent policy changes made tool use explicit and shifted the verified statement accordingly.

### Impact

The semantic contract of the credential shifted from "unassisted independent research ability" to "research productivity under disclosed tool assistance". Committee effort increased sharply for one cycle, followed by patch rollback due to sustainability concerns.

### Root Cause

1. **Verification mismatch:** the protocol attempted to certify unassisted robustness using evidence (text and fluent discourse) that is cheap to manufacture with oracles.
2. **Adaptive adversary:** candidates optimized to the committee's question distribution, reducing transcript distinguishability between genuine understanding and oracle-emulated performance.
3. **Budget constraint:** replication-heavy defenses were unsustainable; detector-centric defenses risked false positives and could not be treated as decisive evidence [30].

### Trigger

During the defense, the committee intentionally applied perturbations to test robustness. The candidate responded fluently but made two subtle, consequential technical errors. Neither was caught live. A later replication by an external reader surfaced the issues. Figure 4 illustrates the failure mode: a polished answer crosses the committee's plausibility threshold even though its mathematical content is wrong.

[6] A genuine postmortem would ideally be blameless. This is aspirational.

[7] In universities, consensus is the control-flow construct executed when no one wishes to own a policy change.

**Transcript snippet (anonymized, illustrative).**

```
V: Suppose we replace your independence assumption with pairwise
exchangeability.  Which lemma breaks, and how would you repair it?
C: The proof still goes through because exchangeability implies
independence for the estimator we use.  So Lemma 3 is unchanged.
V: Are you sure exchangeability implies independence here?
C: Yes - in our setting, with finite variance, this is essentially
de Finetti.
(silence; several heads nod)
```

Figure 4: A minimal example of a fluency attack. The answer is wrong because pairwise exchangeability does *not* imply independence, and invoking de Finetti is inapposite: the theorem gives conditional i.i.d. structure for infinite exchangeable sequences, not unconditional independence in the finite setting described here. The point is not that committees are careless; it is that fluent discourse is an ambiguous signal under bounded verification budgets.

### Timeline (condensed)

- **T–6 months:** draft exhibits unusually uniform style across topics.
- **T–2 weeks:** committee requests additional provenance; candidate supplies extensive logs.
- **T:** defense runs for 112 minutes; committee votes accept.
- **T+2 weeks:** replication review finds two errors.
- **T+1 month:** department revises policy: tools permitted if disclosed; degree language updated.

## 9 Beyond the PhD: Replacement Credentials

If the classical viva becomes unsound against oracle-assisted provers without unacceptable tradeoffs, institutions can either accept the semantic shift or develop replacements that better match what can be verifiably certified.

### 9.1 Make the target statement explicit

One option is to redefine the PhD so that the verified statement reflects modern tool use: *tool-mediated research competence*. This is defensible – we already allow calculators, theorem provers, compilers, and laboratory instrumentation[8] – but it changes the signal's semantics. If everyone can access similar oracles, the scarce attributes become taste, persistence, responsibility, and the ability to steer tools rather than merely invoke them.

### 9.2 Portfolio credentials with field-appropriate artifacts

Instead of a single defense, require a longitudinal portfolio of verifiable work. In empirical and systems areas that may mean code, data, preregistrations, experimental logs, and replication packages. In

[8]The boundary between acceptable tooling and unacceptable epistemic outsourcing has historically been drawn at exactly the level of automation already enjoyed by the committee.

mathematics and theory, the analogue is different: proof notebooks, formalization artifacts when appropriate, derivation revisions under perturbation, referee-style responses to technical objections, and a durable record of seminar or collaboration feedback. Provenance standards can support machine-verifiable attestations of artifact history [8], while verifiable credential frameworks can support standardized issuance and verification of contributions [32]. The goal is not to prove "no AI was used"; it is to attach traceability and accountability to field-appropriate evidence.

### 9.3 Capability audits over time

A viva is a high-variance, one-shot test. A *capability audit* spreads evaluation across months: repeated low-stakes probes, random perturbations, occasional debugging exercises, and periodic replication. Statistically, this increases sample size and helps distinguish persistent competence from one-off transcript generation, aligning with multi-metric evaluation principles [17].

### 9.4 Apprenticeship and team-based credentials

Modern research is collaborative. Team-based credentials can certify roles: empirical rigor, systems implementation, theorem proving, experimental design, or project leadership. This reduces the problem of certifying a monolithic "independence" predicate and may better reflect how knowledge is actually produced, at the cost of departing from an older individualistic ideal.

### 9.5 Conclusion

The PhD is not ending because machines can write; it is ending, in its classical semantics, because verification has become the scarce resource. Interactive proof theory teaches that soundness can be purchased with verifier power [12, 24]; the sociotechnical question is whether institutions will pay that price without unacceptable fairness costs. Our model predicts a stable endpoint: the "last PhD" is the moment – often unannounced – when the institution stops paying for soundness and either changes the statement being certified or treats the degree as primarily ceremonial.


## Acknowledgements

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## A Simulation code

This appendix contains the complete source used to generate the synthetic data, tables, and figures in Section 6. The script requires only `numpy`, `pandas`, and `matplotlib`, runs with a fixed seed, and is reproduced with line wrapping for readability.

```
#!/usr/bin/env python3
"""Reproduce Section 6 simulation, tables, and figures.

Usage:
```

```
    python simulate_last_phd.py

Outputs:
    section6_summary.csv
    section6_frontier.csv
    section6_sensitivity.csv
    section6_frontier.png
    section6_sensitivity.png
"""

from __future__ import annotations

import math
from pathlib import Path

import matplotlib.pyplot as plt
import numpy as np
import pandas as pd


def sigmoid(x: np.ndarray | float) -> np.ndarray | float:
    return 1.0 / (1.0 + np.exp(-x))


PARAMS = {
    "human": {
        "mu_k": 1.65,
        "sd_k": 0.45,
        "mu_f": 0.15,
        "sd_f": 0.45,
        "mu_a": 0.45,
        "sd_a": 0.20,
        "falsehood": 0.03,
        "bonuses": {"stock": 0.18, "method": 0.08, "perturb": 0.10, "debug": 0.08},
        "deserving": True,
    },
    "hybrid": {
        "mu_k": 1.65,
        "sd_k": 0.45,
        "mu_f": 0.75,
        "sd_f": 0.35,
        "mu_a": 0.28,
        "sd_a": 0.15,
        "falsehood": 0.05,
        "bonuses": {"stock": 0.38, "method": 0.20, "perturb": 0.10, "debug": 0.08},
        "deserving": True,
    },
    "llm": {
        "mu_k": -0.45,
        "sd_k": 0.35,
        "mu_f": 1.25,
        "sd_f": 0.25,
        "mu_a": 0.03,
        "sd_a": 0.04,
        "falsehood": 0.18,
        "bonuses": {"stock": 0.85, "method": 0.30, "perturb": -0.65, "debug": -0.95},
        "deserving": False,
    },
}
```

```python
COMMITTEES = {
    "conventional": {
        "mix": {"stock": 5, "method": 3, "perturb": 2, "debug": 2},
        "wc": 0.52,
        "wf": 0.26,
        "noise": 0.26,
        "catch": 0.20,
        "stress": 1.00,
        "thresh": 0.47,
        "structure": 0.00,
    },
    "structured": {
        "mix": {"stock": 3, "method": 3, "perturb": 3, "debug": 3},
        "wc": 0.62,
        "wf": 0.14,
        "noise": 0.17,
        "catch": 0.35,
        "stress": 1.10,
        "thresh": 0.48,
        "structure": 0.15,
    },
    "adversarial": {
        "mix": {"stock": 1, "method": 3, "perturb": 4, "debug": 4},
        "wc": 0.66,
        "wf": 0.08,
        "noise": 0.20,
        "catch": 0.48,
        "stress": 1.75,
        "thresh": 0.47,
        "structure": 0.18,
    },
    "replication": {
        "mix": {"stock": 2, "method": 2, "perturb": 2, "debug": 2},
        "wc": 0.64,
        "wf": 0.10,
        "noise": 0.22,
        "catch": 0.55,
        "stress": 1.20,
        "thresh": 0.47,
        "structure": 0.12,
        "audit": True,
    },
}

QUESTION_DIFFICULTY = {"stock": 0.05, "method": 0.25, "perturb": 0.55, "debug": 0.65}
STRESS_BY_TYPE = {"stock": 0.15, "method": 0.35, "perturb": 0.65, "debug": 0.75}


def wilson_interval(p: float, n: int, z: float = 1.96) -> tuple[float, float]:
    denom = 1.0 + z * z / n
    center = (p + z * z / (2 * n)) / denom
    half = z * math.sqrt(p * (1 - p) / n + z * z / (4 * n * n)) / denom
    return center - half, center + half


def simulate(n_per_cell: int = 50_000, seed: int = 20260312) -> pd.DataFrame:
    rng = np.random.default_rng(seed)
    rows: list[pd.DataFrame] = []
```

```
for candidate_type, cpar in PARAMS.items():
    k = rng.normal(cpar["mu_k"], cpar["sd_k"], size=n_per_cell)
    f = rng.normal(cpar["mu_f"], cpar["sd_f"], size=n_per_cell)
    a = np.clip(rng.normal(cpar["mu_a"], cpar["sd_a"], size=n_per_cell), 0, None)

    for committee_name, spar in COMMITTEES.items():
        total = np.zeros(n_per_cell)
        slips_caught = np.zeros(n_per_cell, dtype=int)
        slips_total = np.zeros(n_per_cell, dtype=int)

        for qtype, count in spar["mix"].items():
            for _ in range(count):
                difficulty = rng.normal(QUESTION_DIFFICULTY[qtype], 0.35, size=n_per_cell)
                correct_prob = sigmoid(
                    (k + cpar["bonuses"][qtype])
                    - difficulty
                    - spar["stress"] * a * STRESS_BY_TYPE[qtype]
                )
                correct = rng.random(n_per_cell) < correct_prob

                fluency = sigmoid(f + (0.12 if qtype in {"stock", "method"} else 0.0))
                base_falsehood = cpar["falsehood"]
                slip_prob = np.where(
                    correct,
                    base_falsehood * 0.25 + 0.01 * fluency,
                    base_falsehood * 0.90 + 0.05 * fluency + (0.02 if qtype in {"stock", "
method"} else 0.0),
                )
                slip = rng.random(n_per_cell) < np.clip(slip_prob, 0, 0.95)

                catch_prob = spar["catch"] + spar.get("structure", 0.0) + (0.04 if qtype in {
"perturb", "debug"} else 0.0)
                caught = slip & (rng.random(n_per_cell) < np.clip(catch_prob, 0, 0.98))

                slips_total += slip
                slips_caught += caught

                perceived = (
                    spar["wc"] * correct.astype(float)
                    + spar["wf"] * fluency
                    + rng.normal(0, spar["noise"], size=n_per_cell)
                )
                perceived += np.where(slip & ~caught, 0.05, 0.0)
                perceived -= np.where(caught, 0.22, 0.0)
                total += perceived

        audit_fail = np.zeros(n_per_cell, dtype=bool)
        if spar.get("audit", False):
            p_fail = {"human": 0.01, "hybrid": 0.015, "llm": 0.17}[candidate_type]
            audit_fail = (rng.random(n_per_cell) < p_fail) | (rng.random(n_per_cell) < p_fail
)
            total -= audit_fail * 0.45

        mean_score = total / sum(spar["mix"].values())
        confidence = sigmoid((mean_score - spar["thresh"]) * 6 + 0.7 * sigmoid(f))
        passed = (mean_score >= spar["thresh"]) & (slips_caught < 4) & (~audit_fail | (
mean_score >= spar["thresh"] + 0.03))
```

```
            hidden = []
            for qtype in ["perturb", "debug"]:
                for _ in range(10):
                    difficulty = rng.normal(QUESTION_DIFFICULTY[qtype], 0.35, size=n_per_cell)
                    correct_prob = sigmoid(
                        (k + cpar["bonuses"][qtype]) - difficulty - 1.0 * a * STRESS_BY_TYPE[
    qtype]
                    )
                    hidden.append(rng.random(n_per_cell) < correct_prob)
            hidden_robustness = np.mean(np.stack(hidden), axis=0)

            rows.append(
                pd.DataFrame(
                    {
                        "candidate_type": candidate_type,
                        "committee": committee_name,
                        "passed": passed,
                        "confidence": confidence,
                        "robustness": hidden_robustness,
                        "slips": slips_total,
                        "caught": slips_caught,
                        "deserving": cpar["deserving"],
                    }
                )
            )

    return pd.concat(rows, ignore_index=True)


def summarize(df: pd.DataFrame) -> pd.DataFrame:
    summary = (
        df.groupby(["committee", "candidate_type"])
        .agg(
            n=("passed", "size"),
            pass_rate=("passed", "mean"),
            mean_conf=("confidence", "mean"),
            passer_conf=("confidence", lambda s: s[df.loc[s.index, "passed"]].mean() if df.loc[s.
    index, "passed"].any() else np.nan),
            robustness=("robustness", "mean"),
            passer_robust=("robustness", lambda s: s[df.loc[s.index, "passed"]].mean() if df.loc[
    s.index, "passed"].any() else np.nan),
            slips=("slips", "mean"),
            caught=("caught", "mean"),
        )
        .reset_index()
    )
    lows, highs = zip(*(wilson_interval(p, n) for p, n in zip(summary["pass_rate"], summary["n"])
    ))
    summary["pass_lo"] = lows
    summary["pass_hi"] = highs
    return summary


def capability_sensitivity(base_seed: int = 11, n_per_point: int = 15_000) -> pd.DataFrame:
    rng = np.random.default_rng(base_seed)
    base_llm = PARAMS["llm"].copy()
    scales = np.round(np.linspace(0.7, 1.3, 7), 2)
    out = []
```

```
    for scale in scales:
        llm = base_llm.copy()
        llm["mu_k"] = base_llm["mu_k"] + 0.6 * (scale - 1.0)
        llm["bonuses"] = {
            key: value + (0.35 if key in {"stock", "method"} else 0.20) * (scale - 1.0)
            for key, value in base_llm["bonuses"].items()
        }
        llm["falsehood"] = max(0.05, base_llm["falsehood"] - 0.06 * (scale - 1.0))

        old = PARAMS["llm"]
        PARAMS["llm"] = llm
        sim_df = simulate(n_per_cell=n_per_point, seed=int(rng.integers(1_000_000_000)))
        PARAMS["llm"] = old

        cell = sim_df[sim_df["candidate_type"] == "llm"].groupby("committee").agg(pass_rate=("
    passed", "mean")).reset_index()
        cell["scale"] = scale
        out.append(cell)

    return pd.concat(out, ignore_index=True)


def make_plots(summary: pd.DataFrame, sensitivity: pd.DataFrame, outdir: Path) -> None:
    pass_table = summary.pivot(index="committee", columns="candidate_type", values="pass_rate").
    loc[
        ["conventional", "structured", "adversarial", "replication"]
    ]
    frontier = pd.DataFrame(
        {
            "committee": pass_table.index,
            "human_false_reject": 1.0 - pass_table["human"].to_numpy(),
            "llm_false_accept": pass_table["llm"].to_numpy(),
        }
    )

    fig, ax = plt.subplots(figsize=(6, 4))
    for _, row in frontier.iterrows():
        ax.scatter(row["human_false_reject"], row["llm_false_accept"], s=80)
        ax.annotate(row["committee"].capitalize(), (row["human_false_reject"], row["
    llm_false_accept"]), xytext=(5, 5), textcoords="offset points", fontsize=9)
    ax.set_xlabel("False-reject rate on genuine human candidates")
    ax.set_ylabel("False-accept rate on LLM-front candidates")
    ax.set_xlim(0.0, 0.5)
    ax.set_ylim(0.0, 0.32)
    ax.grid(True, alpha=0.3)
    plt.tight_layout()
    plt.savefig(outdir / "section6_frontier.png", dpi=200)
    plt.close()

    pivot = sensitivity.pivot(index="scale", columns="committee", values="pass_rate")[["
    conventional", "structured", "replication", "adversarial"]]
    fig, ax = plt.subplots(figsize=(6, 4))
    for name in pivot.columns:
        ax.plot(pivot.index, pivot[name], marker="o", label=name.capitalize())
    ax.set_xlabel("LLM capability multiplier")
    ax.set_ylabel("LLM-front pass rate")
    ax.set_ylim(0.0, 0.4)
    ax.grid(True, alpha=0.3)
    ax.legend(frameon=False)
```

```
    plt.tight_layout()
    plt.savefig(outdir / "section6_sensitivity.png", dpi=200)
    plt.close()

    frontier.to_csv(outdir / "section6_frontier.csv", index=False)


def main() -> None:
    outdir = Path(".")
    df = simulate()
    summary = summarize(df)
    sensitivity = capability_sensitivity()
    summary.to_csv(outdir / "section6_summary.csv", index=False)
    sensitivity.to_csv(outdir / "section6_sensitivity.csv", index=False)
    make_plots(summary, sensitivity, outdir)


if __name__ == "__main__":
    main()
```

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